Quiero establecer una política π

$\pi : S \times A \rightarrow \mathbb{R} \qquad \pi(s,a) = P_r[A_T = a \mid S_T = s]$

¿Y como calculo que tan buena es una política?

- Funcion de valor de estado

$$V^{\pi}(s) = E^{\pi}[R_T \mid S_T = s]$$

$$R_T = r_T + \gamma r_{T+1} + \gamma^2 r_{T+2} + \gamma^3 r_{T+3} + \cdots$$
$$= r_T + \gamma [r_{T+1} + \gamma r_{T+2} + \gamma^2 r_{T+3} + \cdots]$$
$$= r_T + \gamma R_{T+1}$$

$$\boxed{V^{\pi}(s) = \sum_{a \in A_{(s)}} \pi(s,a) \sum_{s' \in S} T(s,a,s')[r(s,a,s') + \gamma \underbrace{V^{\pi}(s')}_{E^{\pi}[R_{T+1} \mid S_{T+1} = s']}]}$$

s_1	s_2 o
s_3	s_4

$\pi_1(s, \rightarrow) = 1 \quad \forall s$

$\pi_2(s, a) = \frac{1}{4} \quad \forall s, \forall a$

$$V^{\pi_1}(s_1) = p(s_1, \rightarrow, s_1)[-1 + \gamma V^{\pi_1}(s_1)] + p(s_1, \rightarrow, s_2)[1] + p(s_1, \rightarrow, s_3)[-1 + \gamma V^{\pi_1}(s_3)] + p(s_1, \rightarrow, s_4)[-1 + \gamma V^{\pi}(s_4)]$$

γ = factor de olvido

$$V^{\pi_1}(s_1) = 0.05[-1 + \gamma^{\pi_1}(s_1) + 0.9 + 0.05(-1 + \gamma V^{\pi_1}(s_3)$$

$$V^{\pi_3}(s_3) = 0.05[-1 + \gamma^{\pi_1}(s_1) + 0.9[-1 + \gamma V^{\pi_1}(s_4)] + 0.05(-1 + \gamma V^{\pi_1}(s_3)]$$

$$V^{\pi_1}(s_4) = 0.9[-1 + \gamma V(s_4)] + 0.05[-1 + \gamma V^{\pi_1}(s_4)] + 0.05[1]$$

$\pi_1 \propto \pi_2$ ssi

$V^{\pi_1}(s) \leq V^{\pi_2}(s) \quad \forall s$

π_1 y π_2

π_3 tal que $\pi_3(s) = \begin{cases} \pi_1(s) \text{ si: } V^{\pi_1}(s) \geq V^{\pi_2}(s) \\ \pi_2(s) \text{ en otro caso} \end{cases} \quad \forall s \in S$

$\pi_1 \leq \pi_3$

$\pi_2 \leq \pi_3$

π^* es una politica optima ssi

$V^{\pi^*}(s) \geq V^{\pi}(s) \quad \forall \pi$

Funcion de valor estado-acción

T = transicion

$Q^{\pi}(s,a) = E^{\pi}[R_T \mid S_T = s, A_T = a]$

$Q^{\pi}(s,a) = \sum_{s \in S} T(s,a,s')[r(s,a,s') + \gamma V^{\pi}(s')]$

$$V^{\pi}(s) = \sum_{a \in A(s)} \pi(s,a)\, Q^{\pi}(s,a)$$

$$\pi^* \to V^* \quad Q^*$$

$$\pi^*(s,a) = \begin{cases} 1 \text{ para } a^* \text{ tal que } a^* = \arg\max_a Q^*(s,a) \\ 0 \text{ en otro caso} \end{cases}$$

$$V^*(s) = \max_a Q^*(s,a)$$

$$\boxed{V^*(s) = \max_a \left(\sum_{s' \in S} T(s, a^*, s') \left[r(s, a^*, s') + \gamma V^*(s') \right] \right)}$$

Base de programacion dinamica

Ecuación de optimalidad de bellmann